



Postdoctoral position: Acoustic metamaterials at MHz frequencies (f/m/d)

Universität Heidelberg

 Bewerbungsfrist: 15.09.2026

 Befristet

 Vollzeit

 Heidelberg

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UNIVERSITÄT
HEIDELBERG
ZUKUNFT
SEIT 1386

Heidelberg University is a comprehensive university with a strong focus on research and international standards. With around 32,200 students and 9,540 employees, including numerous top researchers, it is a globally respected institution that also has outstanding economic significance for the Rhine-Neckar metropolitan region.

The Neuenheimer Feld Campus of Heidelberg University is the largest campus for natural and life sciences in Germany, where the Faculty of Engineering Sciences bridges material sciences, molecular biotechnology and computational sciences. The modern Campus harbors the buildings of the science faculties and institutes, in direct vicinity to the University Hospital Heidelberg, the Max Planck Institute for Medical Research, and the German Cancer Research Center (DKFZ). Strong interactions exist with the Karlsruhe Institute of Technology via the DFG-funded Excellence Cluster “3D Matter Made to Order- (3DMM2O). Heidelberg University invites applications for the following full-time postdoc position to begin as soon as possible.

POSTDOCTORAL POSITION: ACOUSTIC METAMATERIALS AT MHZ FREQUENCIES (F/M/D)

Background:

- The position is funded via the Excellence Cluster “3D Matter Made to Order- (3DMM2O), which aims to establish scalable digital three-dimensional (3D) additive manufacturing across all scales from the molecular to the macroscopic range
- 3D acoustic particle manipulation complements light-based fabrication methods by enabling the assembly of a wider range of materials, including living biological cells, into user-defined constructs. Integrating the acoustic technology with established instrumentation requires new approaches to control sound waves with high precision and in confined volumes
- The position will be embedded within the research group of Dr. Kai Melde (IMSEAM, Heidelberg University) and co-supervised by Prof. Carsten Rockstuhl (KIT) to combine engineering and computational expertise. We further collaborate with Dr. Simon Binder (University of Freiburg) regarding the fabrication process

Your Role:

- Develop & test acoustic metamaterial concepts for Megahertz frequencies
- Conduct numerical simulations of sound fields at the unit cell and device level
- Optimize geometries that enable impedance matching, waveguiding & wave modulation to control sound pressure